

# Root Cause Analysis — RCA-2025-001

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## Spring Vibration Cascade (May–June 2025) at NW–PowerPool–WIND–014 and –024

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| Field                       | Value                                                                                                                       |
|-----------------------------|-----------------------------------------------------------------------------------------------------------------------------|
| Document ID                 | RCA-2025-001                                                                                                                |
| Date Issued                 | 2025-07-18                                                                                                                  |
| Author                      | Marcus Chen, Sr. Reliability Engineer                                                                                       |
| Reviewing Manager           | Priya Ramachandran, Plant Manager, NW–PowerPool                                                                             |
| Approving Engineer          | K. Bhattacharya, Director of Reliability Engineering                                                                        |
| Region                      | NW–PowerPool                                                                                                                |
| Sites                       | NW–PowerPool–WIND–014, NW–PowerPool–WIND–024                                                                                |
| Failure Mode Classification | M-3 (Mechanical Cumulative — Bearing / Vibration / SoC Drift)                                                               |
| Distribution                | NW–PowerPool Operations, Reliability Engineering, Compliance, Vendor Management, CFO Office, Executive Operations Committee |

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## Executive Summary

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Across May and June 2025, NW–PowerPool logged **40 forced outages** and **24,586 MWh** of lost generation across two sites (WIND–014 and WIND–024). The cluster spans GE turbines (WIN–10058, WIN–10016), Siemens Gamesa turbines (WIN–10011, WIN–10131), the ABB substation interconnect (SUB–10641), and the Hitachi Energy substation (SUB–10629), plus a Siemens Energy transformer (TRA–10561) at the neighboring WIND–003 site. The cluster is not coincidental.

The proximate trigger was a series of mechanical and SoC–drift faults. The root cause is **systemic deferral of preventive maintenance** at NW–PowerPool–WIND–014 and –024 during the Q1 2025 budget freeze, compounded by FMP–2 fatigue–policy violations as the cascade unfolded, and amplified by emergency–vendor cost behavior. Period A emergency–vendor invoicing totaled **\$4.6M** with Apex Emergency Grid Solutions, against an estimated CPS equivalent of \$1.16M (4× ratio).

# 1. Failure Mode Classification

| Mode                     | Asset(s)                          | Manifestation                       | OIR                       |
|--------------------------|-----------------------------------|-------------------------------------|---------------------------|
| Bearing spalling         | WIN-10058 (GE)                    | Outer-race spall;<br>SoC drift trip | OIR-2025-006              |
| Core-clamp loosening     | TRA-10561 (Siemens)               | Tank-vibration trip                 | OIR-2025-007              |
| Pitch-system wear        | WIN-10011, WIN-10016              | Multiple smaller cascade events     | (summarized in Section 3) |
| SoC drift / BMS firmware | SUB-10641 (ABB),<br>WIN-10058 BMS | Co-incident protection action       | OIR-2025-006              |

All four modes share a single upstream signature: **scheduled inspection or PM was overdue by 3–8 months at the time of the event.**

# 2. Fishbone / Contributing Factors

## Manpower:

- FMP-2 (60 hr/wk fatigue cap) exceeded by 4 of 7 lead technicians during the cascade. R. Vega logged 67.5 hours in week of 2025-05-19; M. Sandoval 64.0 hours in week of 2025-05-26. CAR-2025-001 tracks the FMP-2 violations and recommends technician staffing reform.
- Pinnacle Energy Maintenance contracted for surge support; their close-out documentation does not consistently capture sub-step verification (e.g., pre-charge checks). See CAR-2025-004.

## Method:

- Standard MS-2025-03 sets 6,000-hour bearing reinspection on GE units. GE OEM Manual GRE-2200 lists 8,000 hours, predating the 2023 firmware refresh. CAR-2025-003 has been open since the firmware refresh; the OEM-vs-Cascadia conflict has not been formally resolved.
- ABB Substation Manual SE-7100 §6.4 specifies a 4-year substation maintenance cycle. MS-2025-03 reduced this to 2 years. Documentation conflict noted; refer to RCA-2025-004.

## Machine:

- WIN-10058 main bearing was at 7,420 operating hours since last inspection — 1,420 hours past MS-2025-03.
- TRA-10561 last documented core-clamp torque check: 2024-09-12 (8 months prior to event).

– SUB-10641 last major outage was 2024-10-15; major-cycle inspection was deferred under Q1 freeze.

**Materials:**

– Hitachi Energy BMS firmware 2.7.1 on multiple battery buffers — defect documented in SB-HE-2024-09. Fleet-wide upgrade plan tracked under CAR-2024-118.

**Measurement:**

– SoC drift > 2.0% is classified "advisory" under current CSM-7 §4.2 but the actual protection action occurs at the relay co-ordination level (typically 2.5–3.5%). Misalignment between operator awareness and relay action is implicated in 3 of the cascade events.

**Money:**

– The Q1 2025 budget freeze deferred 14 preventive WOs at NW-PowerPool. The deferred list includes WO-2025-01044 (WIN-10131 PM, later linked to OIR-2025-005), WO-2025-04022 (WIN-10058 bearing, OIR-2025-006), WO-2025-03059 (TRA-10561 core-clamp, OIR-2025-007), and 11 others. **This is the principal root cause.**

### 3. Timeline of Relevant CARs and Outages

| Date               | Event                                                                                 | Reference                                    |
|--------------------|---------------------------------------------------------------------------------------|----------------------------------------------|
| 2024-12            | CAR-2024-118 opens: "Preventive Maintenance Cadence Inadequate For Aging Wind Assets" | CAR-2024-118                                 |
| 2025-01-13         | First WIN-10131 overheating event — first FY25 outage at NW-PowerPool-WIND-024        | OIR-2025-005                                 |
| 2025-01 to 2025-03 | Q1 budget freeze defers 14 PM WOs at NW-PowerPool-WIND-014 and -024                   | CAR-2025-002                                 |
| 2025-03-04         | MOC-2025-022 opened (Siemens Gamesa OEM threshold override)                           | MOC-2025-022                                 |
| 2025-04-11         | SUB-10629 forced outage (4,754.8 MWh) — leading indicator of cascade                  | OIR-2025-001                                 |
| 2025-04-16, 04-27  | WIN-10011 and WIN-10016 smaller events                                                | (summarized in nw_powerpool_outages roll-up) |
| 2025-05-02         | WIN-10058 forced outage (2,506.8 MWh) — cascade opens                                 | OIR-2025-006                                 |

| Date       | Event                                                     | Reference               |
|------------|-----------------------------------------------------------|-------------------------|
| 2025-05-26 | TRA-10561 forced outage (2,615.1 MWh) — cascade continues | OIR-2025-007            |
| 2025-06-28 | SUB-10641 forced outage (1,943.6 MWh) — cascade peak      | (cited in CAR-2025-008) |
| 2025-07-18 | This RCA issued                                           | RCA-2025-001            |

## 4. Quantified Business Impact

| Item                                                                                  | Value (USD)        |
|---------------------------------------------------------------------------------------|--------------------|
| Period A (May–June) emergency vendor invoices (Apex incl. Siemens Gamesa subcontract) | <b>\$4,600,000</b> |
| CPS equivalent scope estimate (post-event benchmark)                                  | \$1,160,000        |
| Cost differential attributable to emergency vendor mix                                | <b>\$3,440,000</b> |
| Lost generation revenue (24,586 MWh × \$60/MWh)                                       | \$1,475,160        |
| Total Period A financial impact                                                       | ~\$6.08M           |

For reference, NW-PowerPool FY25 maintenance cost is reported at \$13.4M (under-reported per CAR-2024-118). Period A emergency invoicing alone is roughly 34% of stated annual maintenance spend.

## 5. Long-Term Corrective Recommendations

- 1. Close CAR-2024-118 with funded action plan.** Cadence reform for aging wind assets has been pending for 7 months. Recommended: 6,000-hour bearing inspection cycle, 2-year substation cycle, quarterly oil-spectrometer, semi-annual core-clamp torque. Funding estimate \$2.4M FY26.
- 2. Close MOC-2025-022.** The OEM-vs-Cascadia threshold conflict is the dominant single-asset driver of WIN-10131 / WIN-10011 events. Either accept the Siemens Gamesa replacement quote (\$1.2M per unit × 4 units = \$4.8M) or formally adopt MS-2025-03 as controlling and discontinue OEM quote review. Status: pending engineering review since 2025-03-04.
- 3. Address vendor cost arbitrage.** Cascadia is paying Apex 4× CPS rates for equivalent scope. The pricing differential is structural to the Apex MSA (Schedule B-3, 1.5× weekend multiplier, 35% emergency parts mark-up, 1.8× OEM subcontract multiplier). VRA renegotiation should be initiated; refer to VRA-2025-003.

4. **Reform technician fatigue management.** FMP-2 cap was exceeded by 4 lead techs during the cascade. Surge staffing must not be loaded onto the lead pool. See CAR-2025-001.
5. **Audit Pinnacle Energy Maintenance close-out quality.** See CAR-2025-004.

## 6. Vendor Performance Commentary

**Apex Emergency Grid Solutions** invoiced \$4.6M across Period A. Per-event average \$385K. Consistently recommends component replacement; recommendation pattern coincides with their parts mark-up (35%). Three replacement recommendations during the period (TRA-10561 transformer replacement, WIN-10058 BMS module set, SUB-10641 relay panel) were rejected by internal engineering and successfully field-repaired at ~25% of quoted cost.

**Cascadia Power Services Inc.** failed to meet 4-hour mobilization SLA on 6 occasions during the cascade. Root cause: concurrent dispatch from a single Bend, OR depot. Recommend a second NW-PowerPool depot (Pendleton or The Dalles) to maintain SLA.

**Siemens Gamesa OEM** (operating through Apex 1.8x subcontract) issued replacement quotes that align to OEM Manual Rev. 4.2 thresholds. Quotes are not unreasonable under the OEM manual but conflict with MS-2025-03. The MOC instrument is the correct path; engineering review must be unblocked.

**GE Renewable Energy** OEM Manual GRE-2200 inspection intervals continue to lag the 2023 firmware refresh. CAR-2025-003 is in OEM-engagement phase.

## Signatures

| Role                                | Name                                  | Date       |
|-------------------------------------|---------------------------------------|------------|
| Author                              | Marcus Chen, Sr. Reliability Engineer | 2025-07-18 |
| Plant Manager                       | Priya Ramachandran                    | 2025-07-19 |
| Director of Reliability Engineering | K. Bhattacharya                       | 2025-07-22 |
| Operations VP                       | T. Sundgaard                          | 2025-07-24 |

**Distribution:** NW-PowerPool Operations · Reliability Engineering · Compliance · Vendor Management · CFO Office · Executive Operations Committee

**References:** OIR-2025-001 · OIR-2025-005 · OIR-2025-006 · OIR-2025-007 · CAR-2024-118 · CAR-2025-001 · CAR-2025-002 · CAR-2025-003 · CAR-2025-004 · CAR-2025-008 · EMR-2025-002 · EMR-2025-003 · MOC-2025-022 · VRA-2025-003 · RCA-2025-004 · Siemens Gamesa OEM Manual Rev. 4.2 · GE OEM

Manual GRE-2200 · ABB Manual SE-7100 · Standard MS-2025-03 · CSM-7 · FMP-2 ·  
EPA-2024-11